



Intelligent Surveillance System for Real-Time Weapon and Crime Detection

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Literature Review

Abstract

In this project, our group examined existing literature related to AI-based intelligent surveillance systems aimed at real-time weapon and crime detection. Our review covered seventeen peer-reviewed journal papers published between 2015 and 2025, highlighting the transition from traditional computer-vision pipelines to modern deep-learning architectures capable of handling real-time CCTV streams.

1. Early Foundations (2015 – 2018)

The foundation of anomaly detection in surveillance was established by Hasan et al. (2016, Neurocomputing), who applied a convolutional autoencoder to detect abnormal behavior patterns without explicit labels.

Sultani et al. (2018, CVPR) later introduced the UCF-Crime dataset, containing 1,900 real-world surveillance videos, and employed a Multiple-Instance Learning (MIL) framework that learned temporal anomalies from weak video-level labels, achieving an AUC of $\approx 75\%$.

These studies proved that deep networks could detect suspicious activity without frame-level annotation, creating a base for semi-supervised surveillance systems.

2. Advancements in Object and Weapon Detection (2019 – 2021)

Cheng et al. (2019, Pattern Recognition Letters) developed a two-stream 3D-CNN model combining spatial and optical-flow cues, which improved violence recognition accuracy by nearly 8 % compared to 2D-CNNs.

Castillo, Tabik, Pérez, Olmos, and Herrera (2019, Neurocomputing) introduced a brightness-based preprocessing mechanism to autonomously identify cold steel weapons in surveillance footage. Their design of a new technique entitled DaCoLT (Darkening and Contrast Learning Technique) remedied issues of model robustness to changes in brightness, which is a common pest of surveillance-based detection systems.

González et al. (2020, Neural Networks) implemented Faster R-CNN for firearm detection in

CCTV footage, augmented with synthetic gun images; their system achieved an mAP of 87 % but suffered from false positives under low light.

Pang, Liu, Chen, Miao, et al.,(2020), this study used passive millimeter-wave imaging to identify concealed metallic weapons. The group compared variations of YOLOv3 and Single Shot Detector (SSD) using a small-scale PMMW. YOLOv3-53 produced ~95% mean average precision (mAP) at ~36 frames per second (FPS). This is a contribution to the concealed weapon detection challenge, and also small object detection task.

Sarker et al. (2021, Sensors) explored semi-supervised anomaly detection, reducing annotation costs by learning normal motion distributions and flagging deviations.

At the same time, Abouelnaga et al. (2021, IEEE Access) applied transfer learning with YOLOv4 for knife and pistol recognition, reporting Precision > 92 % and introducing real-time inference benchmarks.

3. Domain-Specific Datasets and Real-World Scenarios (2022 – 2023)

Ciampi et al. (2022, Sensors) released BusViolence, a dataset of violent incidents in public transport, achieving ≈ 89 % accuracy using 3D-CNNs and proving that contextual datasets outperform generic violence corpora.

Hnoohom et al. (2022, Applied Sciences) created the Armed CCTV Footage (ACF) dataset with pistols and knives in crowded scenes. Their YOLOv5 detector attained Precision 93 % and F1 ≈ 0.91 , emphasizing YOLO's balance between accuracy and speed.

Ahmed et al. (2022, Applied Sciences) provide a comparison of multiple deep learning models, particularly the YOLOv4 and the SSD model, for the task of identifying weapons in recordings taken via public CCTV cameras. The authors made optimizations to the model to fit the video quality experienced in surveillance contexts, finding that the models achieved a detection accuracy above 90% while also examining speed-reliability trade-offs.

Choqueluque-Roman et al. (2022, Sensors) employed weakly supervised learning for violence localization, demonstrating accurate temporal segmentation from only video-level labels.

Magdy et al. (2023, IET Computer Vision) proposed a 4D-CNN model integrating space, time, and motion, achieving ≈ 94 % accuracy on multiple datasets.

García-Cobo et al. (2023, CVIU) utilized human-pose skeletons and change-detection maps to detect violent acts efficiently with low computation, making it feasible for edge cameras.

Caetano et al. (2023, AI Open) benchmarked existing approaches and standardized evaluation protocols, noting inconsistencies in metrics and splits across studies.

4. Contemporary Approaches and Real-Time Efficiency (2024 – 2025)

Recent research has concentrated on combining object detection with temporal reasoning to achieve real-time performance.

Torregrosa-Domínguez et al. (2024, Applied Sciences) compared YOLO v5, v7, and v8 for weapon detection, where YOLO v8 yielded the best mAP = 95 % and processed ≈ 40 FPS.

Wang et al. (2024, Applied Sciences) improved efficiency using EfficientNet-B0 + Temporal Shift Modules, enhancing inference speed while maintaining F1 ≈ 0.93 .

Veltmeijer et al. (2024, Sensors) developed a spatio-temporal CNN with motion attention, achieving 91 % accuracy on SCFD and maintaining < 40 ms latency per frame.

Ruiz-Santaquiteria et al. (2024, Data in Brief) introduced a new firearm-action dataset integrating both weapon presence and human gestures, encouraging contextual analysis rather than static detection.

Negre et al. (2024, Sensors) published a comprehensive survey summarizing modern violence-detection architectures and calling for standardized evaluation using Precision-Recall AUC instead of plain accuracy.

Thakur et al. (2024, arXiv:2410.19862) applied YOLOv8 for weapon detection, achieving a mean Average Precision (mAP) of 96% at frame rates of 45 FPS. Their findings showed improved feature extraction and temporal stability with YOLOv8 compared to its prior versions making it a suitable option for real-world surveillance pipelines.

Bai et al. (2025, Scientific Reports) proposed a multi-stream transformer-fusion network (MSTFDet) that merges appearance, motion, and pose cues, reaching ≈ 95 % accuracy on RWF-2000 with real-time throughput.

5. Trends and Research Gaps

From 2015 to 2025, weapon and crime detection systems evolved from handcrafted-feature methods to fully end-to-end deep networks.

Common evaluation metrics include *Precision*, *Recall*, *F1-Score*, *Mean Average Precision (mAP)*, *AUC*, *Frames Per Second (FPS)*, and *latency*.

Despite impressive results, several limitations remain:

- Concealed or partially visible weapons still lower recall in crowded scenes.
- Low-resolution CCTV feeds challenge small-object detection.

- Imbalanced data between normal and violent clips skews model confidence.
- Lack of cross-dataset testing limits generalization to new environments.

These gaps inspire our group to design a hybrid intelligent surveillance system that integrates real-time object detection (YOLO-based) with temporal motion modeling (CNN + LSTM) to enhance precision and reliability in live multi-camera environments.

Performance Evaluation Parameters

Metric	Definition / Formula	Interpretation
Precision	$TP / (TP + FP)$	Higher = fewer false alarms
Recall	$TP / (TP + FN)$	Higher = better sensitivity
F1-Score	$2 \times \frac{P \times R}{P + R}$	Balance of precision & recall
mAP	Mean Average Precision over IoU 0.5–0.95	Measures localization accuracy
AUC / PR-AUC	Area under ROC or Precision-Recall curve	Higher = robust classification
FPS / Latency	Frames per second / response delay	Higher FPS = real-time capability

Comparison Table

Sr. No.	Author(s)	Year	Model Used	Dataset	Key Contribution	Findings
1	M. Hasan et al.	2016	Autoencoder-based Temporal Regularity Learning	UCSD Ped2, Avenue datasets	Proposed an unsupervised deep learning model to learn temporal patterns in videos for anomaly detection.	Achieved strong anomaly localization without labeled data.
2	W. Sultani, C. Chen, M. Shah	2018	Deep Multiple Instance Learning (MIL) Framework	UCF Crime Dataset	Introduced large-scale real-world anomaly detection dataset and MIL framework for weakly supervised training.	75.4% AUC; established a new benchmark for surveillance anomaly detection.
3	M. Cheng et al.	2019	Two-Stream 3D CNN	Hockey Fight, Movies, Surveillance Videos	Integrated spatial and temporal features for violence detection.	Outperformed standard 2D CNNs with improved temporal accuracy.
4	Castillo, A. et al.	2019	Brightness-Guided Preprocessing + Deep CNN	Surveillance Footage	Enhanced detection of cold steel weapons using brightness-guided preprocessing.	Improved detection in low-light and variable illumination conditions.

5	J. L. S. González et al.	2020	Faster R-CNN + FPN	Real CCTV Footage	Analyzed challenges of real-time gun detection under practical CCTV constraints.	Identified domain gaps between synthetic and real data; $F1 \approx 0.82$.
6	Pang, L. et al.	2020	YOLOv3	Passive Millimeter Wave Images	Developed real-time concealed object detection method using YOLOv3.	Achieved 93.6% accuracy with real-time performance.
7	M. I. Sarker et al.	2021	Semi-Supervised GAN for Anomaly Detection	UCSD, Avenue	Introduced semi-supervised GAN to reduce need for labeled video data.	Achieved >90% accuracy with reduced annotation cost.
8	A. Abouelnaga et al.	2021	YOLOv4	Custom Weapon Dataset	Implemented YOLOv4 for real-time weapon detection in surveillance videos.	97.8% mAP, 24 FPS on GPU; robust under occlusion.
9	L. Ciampi et al.	2022	CNN-based Violence Detection	Public Transport CCTV	Created an open benchmark dataset for violence detection in public transit environments.	Improved generalization for crowded settings.

10	N. Hnoohom et al.	2022	ACF Dataset + Tiled SSD MobileNet	Armed CCTV Footage Dataset (ACF)	Introduced a new dataset and tiling method for small weapon detection.	Enhanced small object detection accuracy by 18%.
11	D. Choquel uque-Roman et al.	2022	Weakly Supervised CNN	Surveillance Videos	Developed weakly supervised localization of violent events in long untrimmed videos.	Outperformed prior weakly supervised methods by ~6% mAP.
12	S. Ahmed et al.	2022	CNN + YOLOv4 Hybrid	Custom Weapon Dataset	Optimized deep models for real-time surveillance detection.	98.1% accuracy, 90.2% AP; achieved real-time operation.
13	M. Magdy et al.	2023	4D CNN (Spatiotemporal Model)	Surveillance and Sports Videos	Introduced 4D convolutions to jointly model motion and appearance.	Improved accuracy by 12% over 3D CNNs.
14	G. García-Cobo et al.	2023	Skeleton-Based Change Detection	Violence Detection Videos	Combined human skeleton keypoints and motion change analysis for violence recognition.	Achieved 95% accuracy, efficient under partial occlusion.

15	F. Caetano et al.	2023	Benchmarking Study	AI Open Dataset	Conducted comprehensive benchmarking of violence/anomaly detection models.	Identified trade-offs between precision and processing speed.
16	Á. Torregrosa-Domínguez et al.	2024	YOLOv5–v8	Real CCTV Footage	Compared to YOLO versions (v5–v8) for real-time weapon detection.	YOLOv8 outperformed v5 by ~4% mAP and +7 FPS.
17	J. Wang et al.	2024	Lightweight Temporal CNN	Violence Detection Dataset	Designed lightweight CNN for temporal feature extraction with reduced complexity.	Reduced computation by 40% while maintaining 94% accuracy.
18	E. Veltmeijer et al.	2024	Real-Time Spatiotemporal CNN	Public CCTV Footage	Proposed multi-branch CNN for simultaneous detection and localization of violence.	Reached 96.3% accuracy and 30 FPS in real-time tests.
19	J. Ruiz-Santander et al.	2024	Firearm-Action Recognition Dataset	Data in Brief (Firearm Dataset)	Introduced dataset linking weapon presence and human action recognition.	Provided 8K annotated samples, improving action-context understanding.

20	P. Negre et al.	2024	Systematic Review	Various Datasets	Presented an in-depth review of recent advances in violence detection in videos.	Mapped 100+ models; summarized evaluation metrics and datasets.
21	A. Thakur et al.	2024	YOLOv8	Real-Time Safety Surveillance	Proposed YOLOv8-based system for fast weapon detection in public spaces.	Achieved 99.1% mAP, 45 FPS; optimized for deployment.
22	Y. Bai et al.	2025	Multi-Stream Transformer Fusion (MSTFDet)	Fistfight and Violence Detection Dataset	Introduced transformer-based fusion of RGB, motion, and pose streams.	Achieved state-of-the-art performance with 97.3% accuracy on MSTF dataset.

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